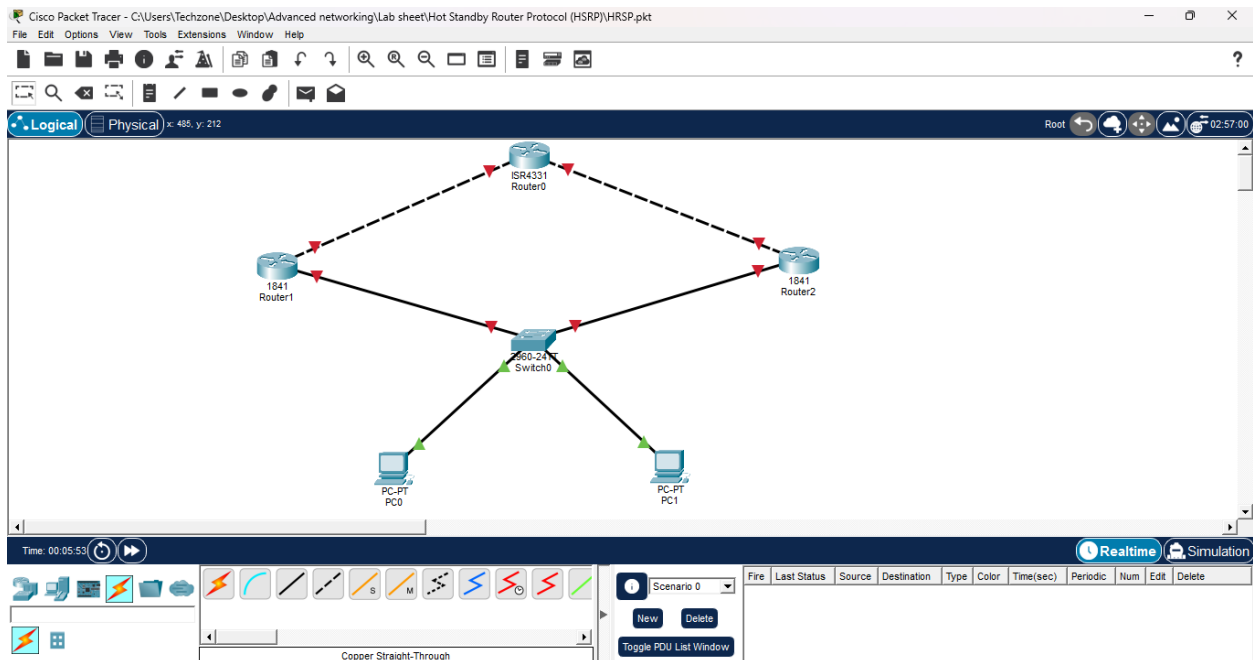


## HSRP



Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
cisco ISR4331/K9 (1RU) processor with 1795999K/6147K bytes of memory.
Processor board ID FIM232010G0
3 Gigabit Ethernet interfaces
32768K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
3207167K bytes of flash memory at bootflash:.
0K bytes of WebUI ODM Files at webui..

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#con ter
% Ambiguous command: "con ter"
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gig0/0/0
Router(config-if)#ip addr 192.168.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

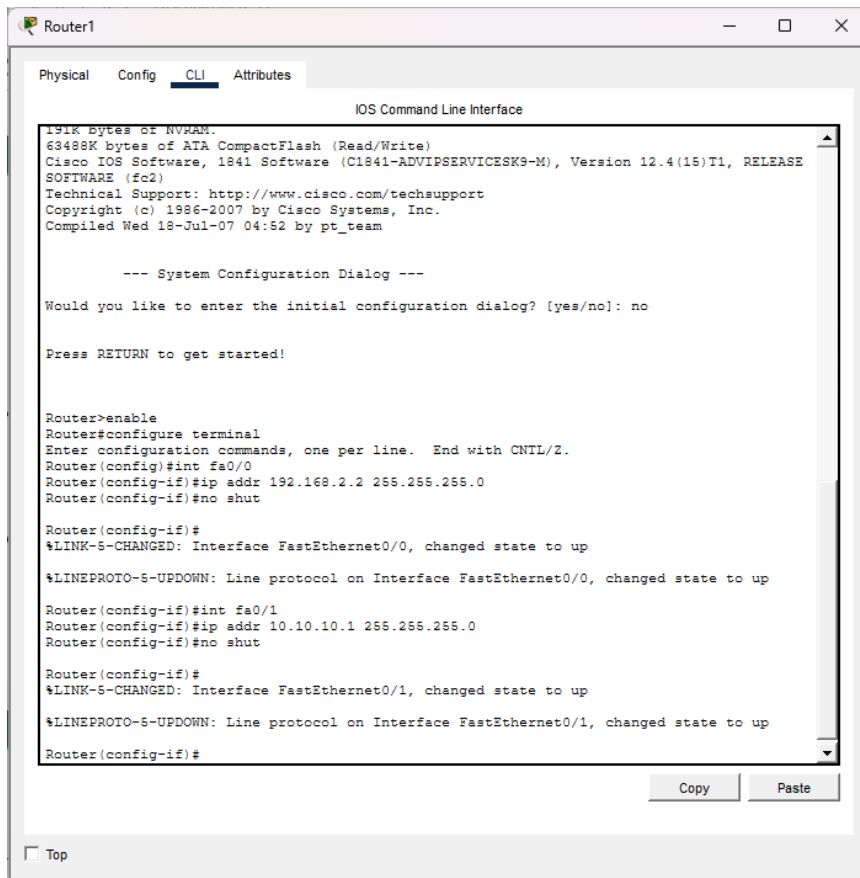
Router(config-if)#int gig0/0/1
Router(config-if)#ip addr 192.168.3.1 255.255.255.0
Router(config-if)#no shut

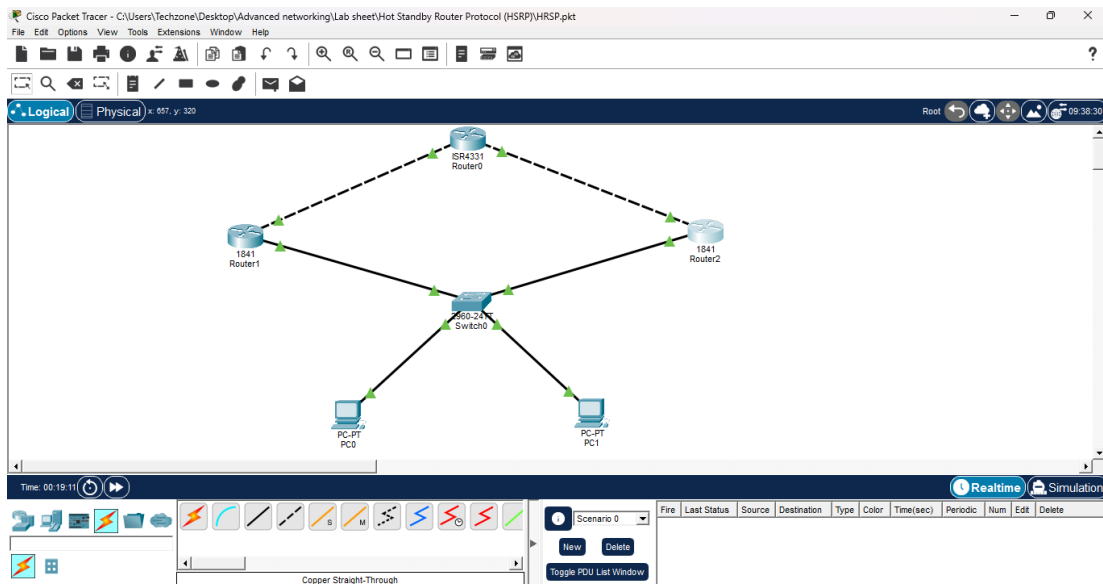
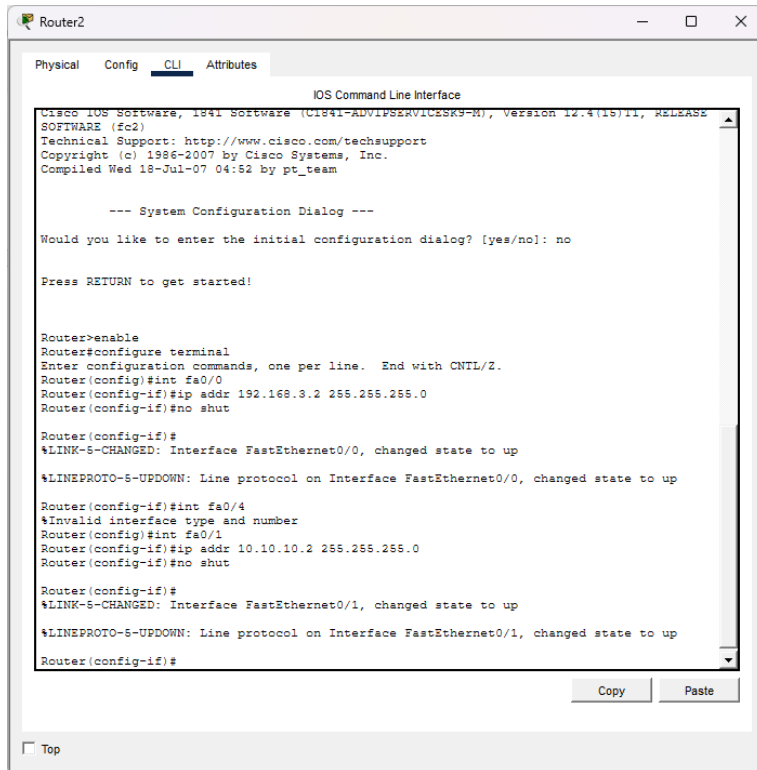
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

Router(config-if)#
```

Copy Paste

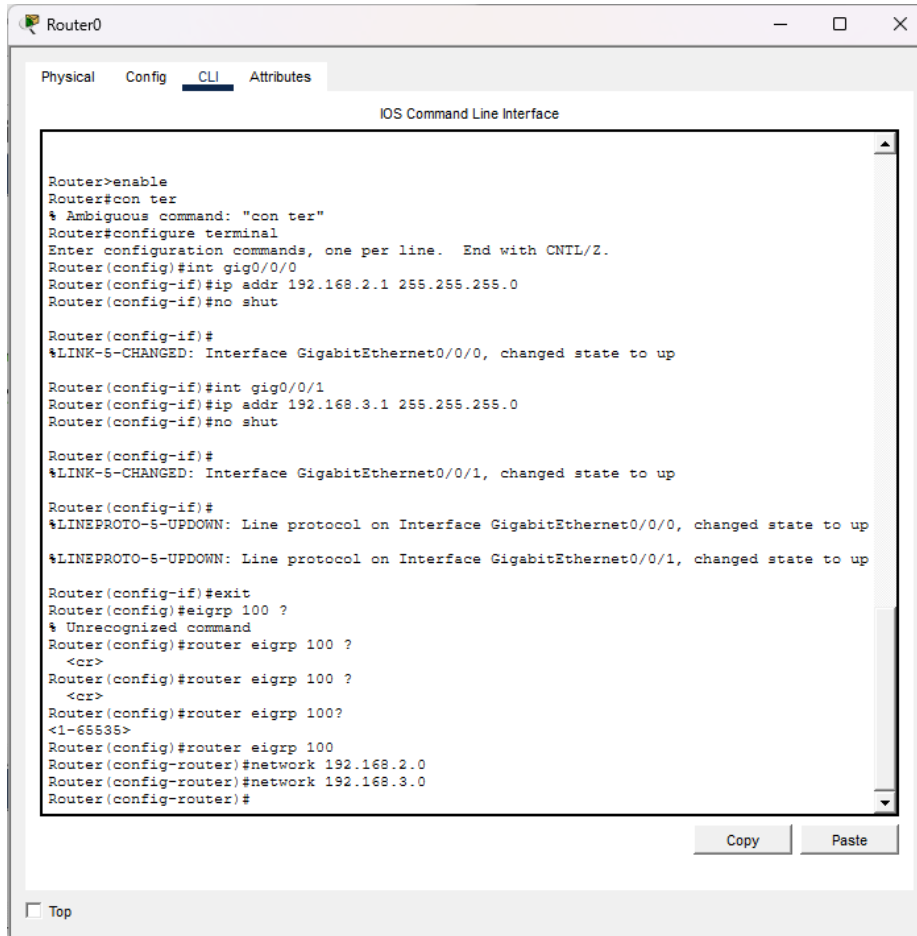
Top





Configure EIGRP on ISR4331 Router, ISP1 and ISP 2. Use 100 as autonomous system number

## ISR4331 ROUTER



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router>enable
Router#con ter
% Ambiguous command: "con ter"
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gig0/0/0
Router(config-if)#ip addr 192.168.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#int gig0/0/1
Router(config-if)#ip addr 192.168.3.1 255.255.255.0
Router(config-if)#no shut

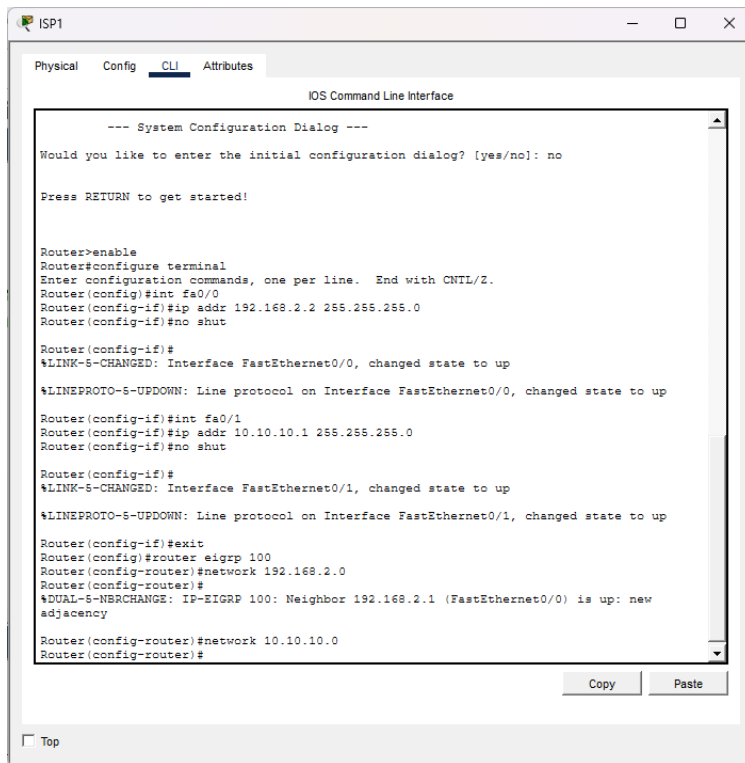
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

Router(config-if)#exit
Router(config)#eigrp 100 ?
% Unrecognized command
Router(config)#router eigrp 100 ?
<cr>
Router(config)#router eigrp 100 ?
<cr>
Router(config)#router eigrp 100?
<1-65535>
Router(config)#router eigrp 100
Router(config-router)#network 192.168.2.0
Router(config-router)#network 192.168.3.0
Router(config-router)#

Copy Paste
Top
```

## ISP1



The screenshot shows a window titled "ISP1" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The interface shows a "System Configuration Dialog" asking to enter the initial configuration dialog, which is answered "no". It then prompts to press RETURN to get started. The configuration process begins with the "enable" command, followed by "configure terminal". The user enters configuration commands for interfaces fa0/0 and fa0/1, including IP addresses and EIGRP settings. The interface status changes to "up" for both. The configuration ends with the "exit" command, and the router returns to the "config-router" prompt.

```
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no

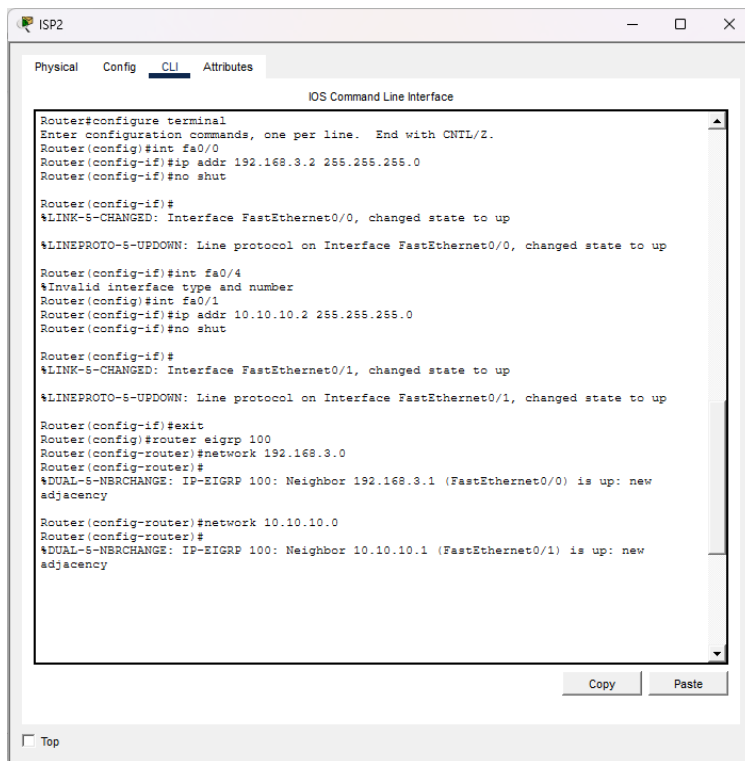
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip addr 192.168.2.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#int fa0/1
Router(config-if)#ip addr 10.10.10.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#router eigrp 100
Router(config-router)#network 192.168.2.0
Router(config-router)#
%DUAL-5-NBCHANGE: IP-EIGRP 100: Neighbor 192.168.2.1 (FastEthernet0/0) is up: new
adjacency
Router(config-router)#network 10.10.10.0
Router(config-router)#
```

## ISP2



The screenshot shows a window titled "ISP2" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The interface shows the configuration process for interfaces fa0/0 and fa0/1, including IP addresses and EIGRP settings. The interface status changes to "up" for both. The configuration ends with the "exit" command, and the router returns to the "config-router" prompt.

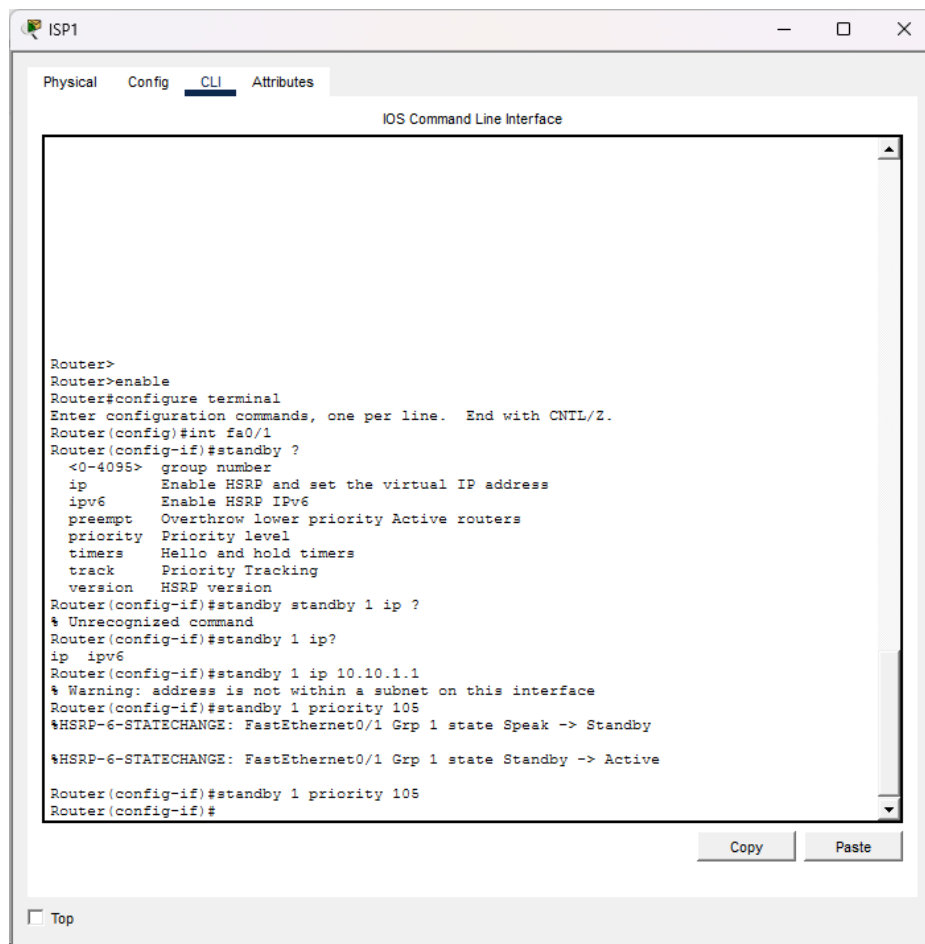
```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip addr 192.168.3.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#int fa0/4
%Invalid interface type and number
Router(config)#int fa0/1
Router(config-if)#ip addr 10.10.10.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#router eigrp 100
Router(config-router)#network 192.168.3.0
Router(config-router)#
%DUAL-5-NBCHANGE: IP-EIGRP 100: Neighbor 192.168.3.1 (FastEthernet0/0) is up: new
adjacency
Router(config-router)#network 10.10.10.0
Router(config-router)#
%DUAL-5-NBCHANGE: IP-EIGRP 100: Neighbor 10.10.10.1 (FastEthernet0/1) is up: new
adjacency
```

We'll configure ISP1 as Active and ISP 2 as Standby. Use group number as 1 and virtual Ip as 10.10.1.1. For ISP1 make the priority as 105 and for ISP2 priority as 100. Thus, the router with the highest priority is elected the active router –ISP 1 in the lab example. The router with the second highest priority becomes the standby router – ISP 2 in the lab example. If all priorities are equal, whichever router has the highest IP Address on its HSRP interface is elected as the active router.

## ISP1 config

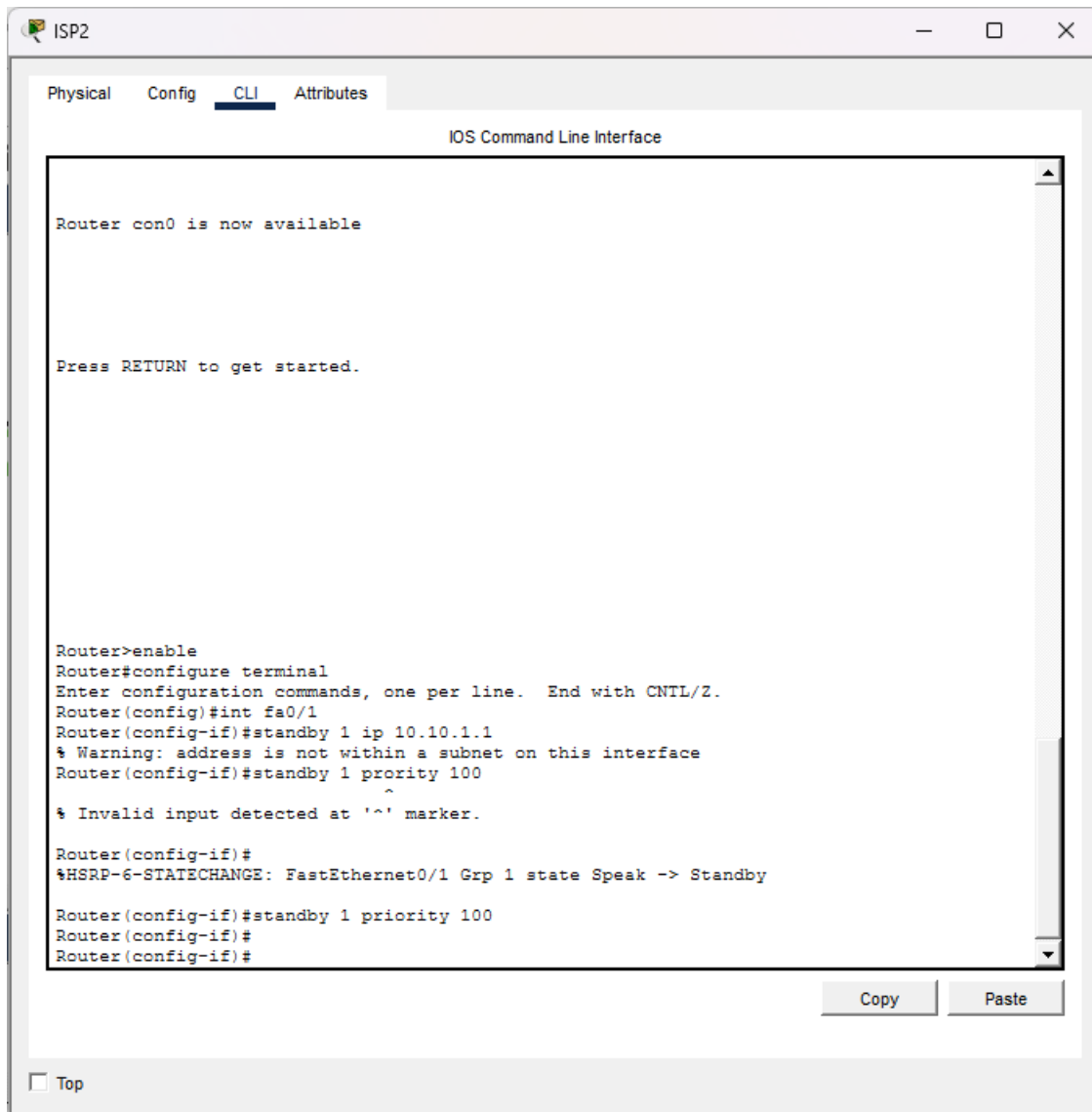


The screenshot shows a window titled "ISP1" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The terminal output shows the following commands and responses:

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/1
Router(config-if)#standby ?
  <0-4095> group number
  ip      Enable HSRP and set the virtual IP address
  ipv6    Enable HSRP IPv6
  preempt Overthrow lower priority Active routers
  priority Priority level
  timers  Hello and hold timers
  track   Priority Tracking
  version HSRP version
Router(config-if)#standby standby 1 ip ?
% Unrecognized command
Router(config-if)#standby 1 ip?
ip ipv6
Router(config-if)#standby 1 ip 10.10.1.1
% Warning: address is not within a subnet on this interface
Router(config-if)#standby 1 priority 105
%HSRP-6-STATECHANGE: FastEthernet0/1 Grp 1 state Speak -> Standby
%HSRP-6-STATECHANGE: FastEthernet0/1 Grp 1 state Standby -> Active
Router(config-if)#standby 1 priority 105
Router(config-if)#
```

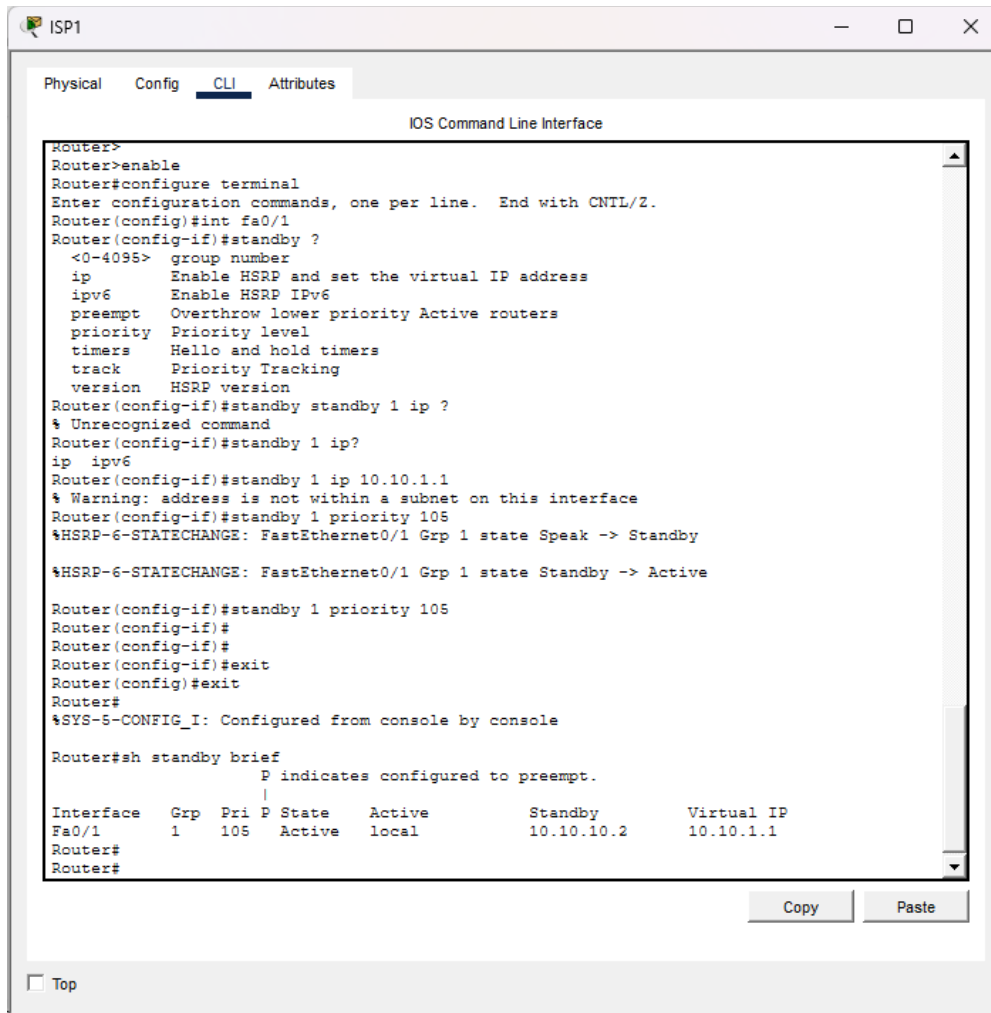
At the bottom of the window, there are "Copy" and "Paste" buttons, and a "Top" button with a checkbox.

## ISP2 config



Run show standby brief command and observe the results

## ISP1



The screenshot shows a Cisco IOS Command Line Interface window titled "ISP1". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The CLI shows the following sequence of commands and outputs:

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/1
Router(config-if)#standby ?
  <0-4095> group number
  ip      Enable HSRP and set the virtual IP address
  ipv6    Enable HSRP IPv6
  preempt Overthrow lower priority Active routers
  priority Priority level
  timers  Hello and hold timers
  track   Priority Tracking
  version HSRP version
Router(config-if)#standby standby 1 ip ?
% Unrecognized command
Router(config-if)#standby 1 ip?
ip ipv6
Router(config-if)#standby 1 ip 10.10.1.1
% Warning: address is not within a subnet on this interface
Router(config-if)#standby 1 priority 105
%HSRP-6-STATECHANGE: FastEthernet0/1 Grp 1 state Speak -> Standby
%HSRP-6-STATECHANGE: FastEthernet0/1 Grp 1 state Standby -> Active

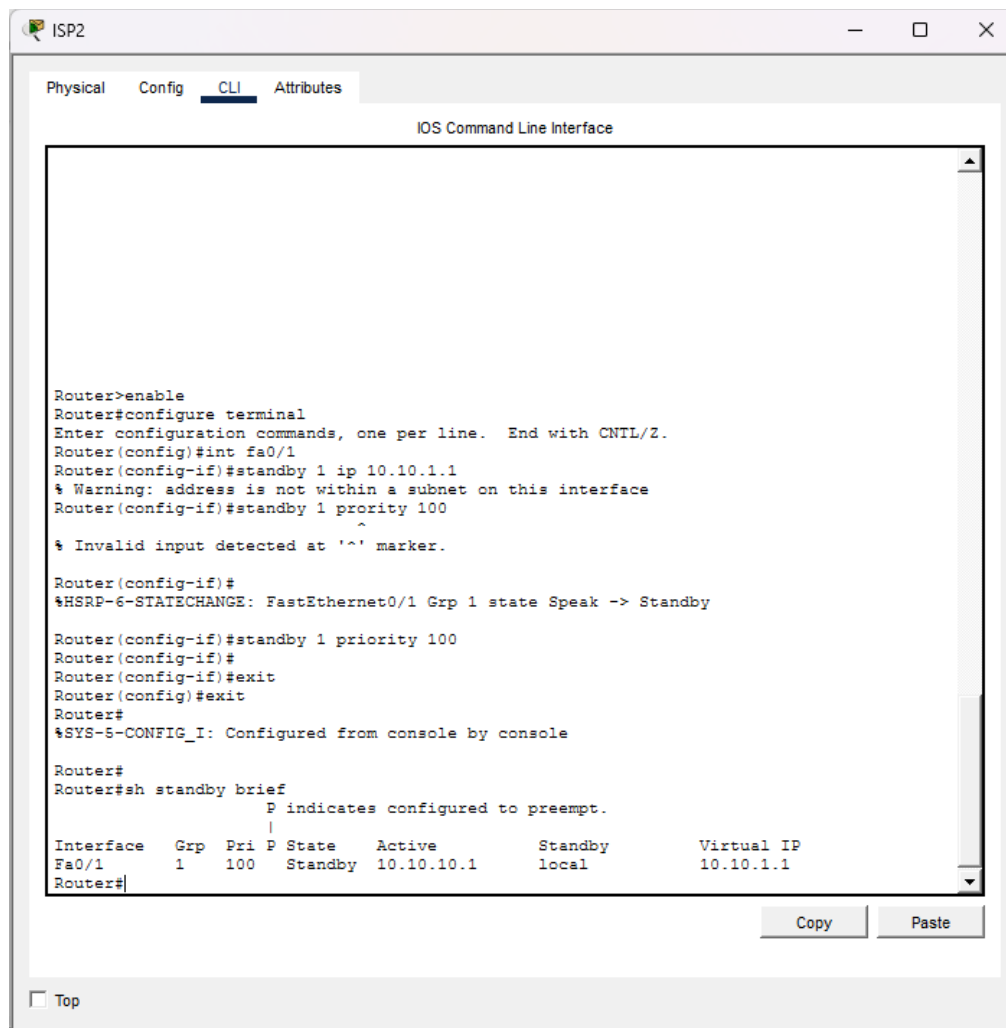
Router(config-if)#standby 1 priority 105
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State   Active      Standby      Virtual IP
Fa0/1      1   105  P Active local    10.10.10.2   10.10.1.1
Router#
Router#
```

At the bottom of the CLI window, there are "Copy" and "Paste" buttons. Below the CLI window, there is a checkbox labeled "Top".



## ISP2



The screenshot shows a terminal window titled "ISP2" with tabs for Physical, Config, CLI (selected), and Attributes. The main area is titled "IOS Command Line Interface". The terminal output shows the following sequence of commands and responses:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/1
Router(config-if)#standby 1 ip 10.10.1.1
% Warning: address is not within a subnet on this interface
Router(config-if)#standby 1 priority 100
Router(config-if)#
% Invalid input detected at '^' marker.
Router(config-if)#
%HSRP-6-STATECHANGE: FastEthernet0/1 Grp 1 state Speak -> Standby
Router(config-if)#standby 1 priority 100
Router(config-if)#
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#sh standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State      Active        Standby        Virtual IP
Fa0/1      1    100  Standby  10.10.1.1    local          10.10.1.1
Router#
```

At the bottom of the terminal window, there are "Copy" and "Paste" buttons. Below the terminal window, there is a checkbox labeled "Top".

Configure client pc as follows and issue a ping command to check connectivity.

The screenshot shows a configuration window for PC0 with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying the IP Configuration settings for the FastEthernet0 interface. The IP Configuration section has two radio buttons: DHCP (unselected) and Static (selected). The Static configuration fields are filled with the following values: IPv4 Address (10.10.10.3), Subnet Mask (255.255.255.0), Default Gateway (10.10.1.1), and DNS Server (0.0.0.0). The IPv6 Configuration section also has two radio buttons: Automatic (unselected) and Static (selected). The Static configuration fields are empty, except for the Link Local Address which is pre-filled with FE80::201:64FF:FE57:C2D5. The 802.1X section has a checkbox for Use 802.1X Security (unchecked), and fields for Authentication (MD5), Username, and Password.

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 10.10.10.3

Subnet Mask: 255.255.255.0

Default Gateway: 10.10.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:64FF:FE57:C2D5

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

